

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

ANNEXURE A

ARTICLE REVIEW

Selected Peer-Reviewed Research Article

Selected article: Ahmed, N., Ahammed, R., Islam, M. M., Uddin, M. A., Akhter, A., Talukder, M. A., & Paul, B. K. (2021). “Machine learning based diabetes prediction and development of smart web application.” *International Journal of Cognitive Computing in Engineering*, 2, 229–241. DOI: 10.1016/j.ijcce.2021.12.001.

The article is a refereed journal article published by Elsevier/KeAi in 2021 and is open access. It directly relates to the CO 4 topics of inductive learning, Decision Trees, Naïve Bayes, Logistic Regression and SVM. ✂cite source omitted in document📞

DOI link: <https://doi.org/10.1016/j.ijcce.2021.12.001>

TASK 1: ARTICLE SUMMARY

1(a) Citation Details, Domain and ML Problem

Full citation: Nazin Ahmed, Rayhan Ahammed, Md. Manowarul Islam, Md. Ashraf Uddin, Arnisha Akhter, Md. Alamin Talukder, and Bikash Kumar Paul, “Machine learning based diabetes prediction and development of smart web application,” *International Journal of Cognitive Computing in Engineering*, Volume 2, June 2021, pp. 229–241, DOI: 10.1016/j.ijcce.2021.12.001.

Domain/Application: Healthcare, medical diagnosis, diabetes risk prediction and clinical decision support.

Specific ML problem: Binary classification of patients into diabetic and non-diabetic classes using clinical and lifestyle-related attributes. The study compares seven supervised learning algorithms—Naïve Bayes, Decision Tree, Random Forest, SVM, Logistic Regression, Gradient Boosting and KNN—and uses the best-performing model to build a Flask-based web application.

1(b) Key Objective and Research Gap

The main objective of the article is to develop a practical machine-learning framework for early diabetes prediction and connect the prediction model to a usable web application. The authors identify a gap in earlier studies: many existing approaches were trained on a single dataset, used limited preprocessing, or reported strong performance only for a particular dataset. This made it difficult to determine whether a model would remain effective across different clinical datasets. To address this gap, the researchers evaluate seven supervised classification methods across multiple diabetes datasets and apply a common preprocessing pipeline involving missing-value handling, outlier removal, label encoding and normalization. They also investigate feature relationships and reduce the input variables to identify important diabetes risk factors. Their experiments show that model performance depends strongly on the dataset and preprocessing strategy. Finally, the authors select a high-performing classifier and integrate it into a Flask web application so that users can enter

patient information and receive a diabetes prediction. The work therefore connects algorithm comparison, preprocessing, feature selection and practical deployment.

TASK 2: METHODOLOGY ANALYSIS

2(a) ML Techniques and Dataset

The article evaluates seven supervised learning techniques: Naïve Bayes (NB), Decision Tree (DT), Random Forest (RF), Support Vector Machine (SVM), Logistic Regression (LR), Gradient Boosting (GB), and K-Nearest Neighbors (KNN). The models are trained to classify diabetes outcomes.

Dataset 1 is the Pima Indian Diabetes Dataset. It contains 768 records and 9 attributes, including Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree Function, Age and the Outcome class. The dataset originates from the National Institute of Diabetes and Digestive and Kidney Diseases.

Dataset 2 contains 950 records and 19 attributes. It includes Age, Gender, Family Diabetes History, high blood pressure, physical activity, BMI, smoking, alcohol consumption, sleep, medicine use, junk-food consumption, stress, blood-pressure level, pregnancies, previous diabetes, urination frequency and the diabetes outcome.

Preprocessing included removal of outliers using the Interquartile Range (IQR) method, handling missing values using feature means, label encoding of categorical values, and standardization/normalization. Correlation analysis was also used to identify influential attributes. For the Pima dataset, glucose, age, BMI and pregnancies were reported as strongly associated with the outcome; five variables—Pregnancies, Glucose, BMI, Diabetes Pedigree Function and Age—were then selected for a reduced-feature experiment.

For evaluation, the study used 10-fold cross-validation, where the dataset is randomly divided into ten parts and each part is used as a test fold once while the remaining folds are used for training.

2(b) Mapping to CO 4, Strength and Limitation

CO 4 mapping:

- Inductive Learning: The models learn general classification patterns from labelled patient records and apply them to unseen patient records.
- Decision Trees: The study directly evaluates a Decision Tree classifier and Random Forest, which is an ensemble of decision trees.
- Statistical Learning: Naïve Bayes and Logistic Regression are included as statistical classification methods, while SVM represents another supervised statistical learning approach.
- Reinforcement Learning: Reinforcement Learning is not used in the article; the work is entirely based on supervised learning.

Methodological strength: A major strength is the comparison of several algorithms across multiple datasets combined with a consistent preprocessing pipeline. This provides a broader view of model performance than testing only one algorithm on one dataset.

Methodological limitation: The datasets are relatively small and do not provide strong evidence of external clinical validity. In addition, very high performance on a specific dataset may not generalize to patients from different hospitals, regions or demographic groups. The study also emphasizes accuracy, whereas clinical screening should additionally prioritize sensitivity, calibration, external validation and the consequences of false negatives.

TASK 3: RESULTS & CRITICAL EVALUATION

3(a) Key Results

For the Pima Indian dataset using all features, the reported accuracies were: NB 78.95%, DT 76.32%, RF 80.26%, SVM 80.26%, LR 77.63%, GB 78.95% and KNN 75.00%. The article reports that SVM performed best in the corresponding overall experiment, while a reduced five-feature experiment achieved 81.13% accuracy with SVM.

For Dataset 2, the reported accuracies were: NB 86.17%, DT 96.81%, RF 96.81%, SVM 91.49%, LR 84.04%, GB 90.43% and KNN 90.43%. The authors report that DT and RF had the highest accuracy, while RF gave the stronger ROC-curve performance.

The paper also reports that, after preprocessing and feature selection, the proposed results improved over selected earlier studies. The final web application used an SVM-based model for the deployment example.

Model	NB	DT	RF	SVM	LR	GB	KNN
Pima Accuracy	78.95%	76.32%	80.26%	80.26%	77.63%	78.95%	75.00%
Dataset 2 Accuracy	86.17%	96.81%	96.81%	91.49%	84.04%	90.43%	90.43%

3(b) Critical Evaluation

The reported results are plausible for controlled benchmark datasets, but the very high 96.81% accuracy on Dataset 2 should be interpreted cautiously. Accuracy can be misleading if class proportions are uneven or if the evaluation protocol does not reflect deployment conditions. The study's use of 10-fold cross-validation is stronger than relying on a single train-test split, but repeated cross-validation, confidence intervals and a completely independent external test set would provide stronger evidence.

Potential biases include dataset-specific sampling, demographic limitations, limited geographic diversity, and possible overfitting to commonly used benchmark datasets. The Pima dataset is restricted to women of Pima Indian heritage, so its results cannot automatically be generalized to all populations. Dataset 2 is also from a specific population and contains self-reported lifestyle variables, which can introduce measurement or reporting bias.

Additional experiments that would strengthen the findings include external validation on hospital data from a different region, repeated stratified cross-validation, calibration analysis, confidence intervals, sensitivity/specificity and precision-recall analysis, subgroup fairness analysis, and

explainability using SHAP or permutation importance. A prospective clinical study would be needed before real clinical deployment.

3(c) Real-World Applicability

The approach could be deployed as a screening or clinical decision-support tool in primary-care clinics, diabetes screening camps, hospital portals, telemedicine systems and mobile/web health applications. A web interface can allow healthcare workers to enter patient measurements and receive a risk classification.

Industry scaling challenges include access to sufficiently large and representative clinical datasets, interoperability with electronic health-record systems, data privacy, model monitoring, computational and infrastructure requirements, changing disease patterns, explainability and regulatory approval. A model trained on public datasets should not be treated as a clinically validated diagnostic device without external and prospective validation.

Ethically, the system should support clinicians rather than replace them. Patient consent, privacy protection, fairness across demographic groups, transparent explanations and mechanisms for human override are important requirements.

TASK 4: REFLECTION & LEARNING OUTCOME

4(a) Three Key Insights

1. Preprocessing can materially change model performance. The article demonstrates that missing-value treatment, outlier handling, encoding and normalization are not merely technical steps; they can directly affect predictive accuracy. This connects to the CO 4 concept of data preparation in inductive learning.
2. Algorithm performance is dataset-dependent. Decision Trees and Random Forests performed extremely well on the second dataset, whereas SVM was competitive on the Pima dataset. This reinforces the CO 4 concept that there is no universally best classifier and that model selection must be empirical.
3. Feature selection improves interpretability as well as efficiency. The identification of glucose, BMI, age and pregnancy-related factors demonstrates how statistical relationships can guide model simplification. This connects to CO 4 topics on statistical learning, feature analysis and interpretable classification.

4(b) Proposed Extension / Novel Experiment

A strong extension would be to build an externally validated, explainable ensemble model using a larger multi-hospital diabetes dataset. The experiment could compare the original models with calibrated Random Forest/XGBoost models and Logistic Regression, followed by SHAP-based explanation and subgroup fairness analysis.

This is feasible because the existing article already provides a preprocessing and multi-model evaluation framework. The main enhancement would be adding a geographically and demographically diverse dataset, external testing and explainability. The expected impact is improved

generalizability and greater trustworthiness for clinical decision support. Instead of optimizing only accuracy, the extended study could optimize a clinically meaningful combination of recall, specificity, AUC-ROC, calibration and fairness.

REFERENCE

Ahmed, N., Ahammed, R., Islam, M. M., Uddin, M. A., Akhter, A., Talukder, M. A., & Paul, B. K. (2021). Machine learning based diabetes prediction and development of smart web application. *International Journal of Cognitive Computing in Engineering*, 2, 229–241. <https://doi.org/10.1016/j.ijcce.2021.12.001>

Source verification: The article is listed as a refereed journal contribution and the publisher record identifies it as an open-access 2021 article in the *International Journal of Cognitive Computing in Engineering*, Volume 2, pages 229–241. The reported dataset sizes, features, preprocessing methods and model accuracies were checked against the article text and tables. ✂cite source omitted in document📞